Measure missingness where the statistic lives: a continuous screening law for gap-affected stream-temperature records

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Key points

- Whether a data gap matters depends on how much the target statistic depends on the days it hides. Sweeping the width of the statistic's support from one day to a full year, the governing axis turns over smoothly from *when* the outage fell to *how much* of the record it removed, and the crossover is at a support of **91 days**.
- The rule that follows is deployable and it changes real decisions: on 2,422 genuinely incomplete station-years from nine agencies, screening on the station's climatological peak window rather than on gap length reverses the admission decision for **15.1%** of them, and removes the 17 station-years that published completeness guidance admits with the peak window entirely unobserved.
- Screening is not the only lever and not the largest. Real outages of three months or more begin in autumn and winter, which protects the summer peak by accident; moving maintenance into spring would cost **4.4 percentage points** of consequential error and moving it into autumn would save **3.9**, against a screening gain of 6.3.

Abstract

Thermal archives are screened before use, almost always by a threshold on record completeness: a minimum share of the year, or a maximum consecutive gap. Both thresholds measure the *size* of an outage, and both are applied to whichever statistic an analyst happens to want. We show that the relevant quantity is neither the size nor the timing of an outage as such, but the share of the target statistic's own support that the outage hides, and that the familiar contrast between "timing" and "length" is the two ends of one continuous scale. On 5,114 fully observed river station-years from 146 networks and nine agencies, with 51,524 outage realizations, we sweep the support of the estimand from a single day to the whole year: the advantage of a peak-window axis over a whole-year completeness axis decays from +0.074 AUC and crosses zero at a support of 91 days, after which the completeness axis wins by up to 0.173. The same law reproduces, on the three sine-fit parameters used by the standing guidance, the pattern that guidance reported as inconsistent. Because a statistic's support has to be located before it can be conditioned on, and its location wanders between years by a median of 7 to 17 days, the conditioning set that performs best is the support *dilated* by that wander: one climatological window of 60 to 90 days is near-optimal for every endpoint from the annual maximum to the half-year mean. Acting on the law, a pre-registered screen scored once on 36 reserved networks cuts the consequential-error rate from 9.7% to 2.5% at equal retention. We then take the rule to the part of the archive the study cannot score: on 2,422 real incomplete station-years it reverses the admission decision for 15.1%, and the published guideline it replaces admits 17 station-years whose annual peak was never observed. Two boundaries are reported as prominently: under the outage geometry these networks actually experience the screening gain is not resolvable, because real long outages begin in autumn; and a screen that selects on difficulty selects against high-elevation, donor-poor catchments, which costs 0.31 points of error to correct and should be corrected.

1. Introduction

Stream temperature is measured to answer questions about thermal regime: was this summer stressful for cold-water fish, is the annual peak rising, how many years exceeded a criterion. Those questions are answered from derived statistics computed on a completed calendar year, and the records they are computed from have holes in them. Loggers fail, are buried by ice, are lost to floods, or are simply not visited. Before a station-year enters an analysis, an analyst must decide whether its statistic can be trusted.

That decision is made today with a completeness rule, and the rules in use are of two kinds. Agency monitoring standards specify what must be present: Mauger et al. (2015), writing minimum standards for a state whose records feed regional analyses, require a minimum sampling duration of one calendar month and recommend a deployment period of 1 June to 31 August "to capture the thermal maxima". Published quantitative guidance instead constrains what may be absent: Johnson et al. (2021) report that "datasets with less than seven weeks of consecutively missing data enabled estimation of all annual signal parameters with reasonable accuracy" and that "imputation of missing data expanded this range to approximately

20 weeks". Analysts then apply thresholds of this second kind to whatever statistic they want. Mauger et al. (2017), for example, retain a site-year for a mean July temperature only "when at least 90% of the days were captured", and then report the maximum weekly maximum temperature from the same records. Neither kind of rule has been tested against an alternative on a large panel where the truth is known.

Two things are worth saying plainly about that guidance before we depart from it. Johnson et al. estimate a *sine-fit* mean, phase and amplitude; they do not compute a hottest 30-day mean and they make no claim about warm-season extremes. And the Alaska standards already condition on season rather than on size, which is closer to what we will argue for than a completeness threshold is. The target of this paper is therefore not the guidance but the habit of carrying a completeness threshold across to an estimand it was not derived for.

The estimands do not depend on the record in the same way. An annual mean is a sum over days: removing k days removes k terms and it matters little which. A hottest 30-day mean is an extreme living in a specific window: if the outage misses that window the statistic is recoverable almost exactly however long the outage is, and if the outage covers it the statistic must be inferred however short the outage is. Stated that way the contrast looks like a dichotomy between timing and length, and that is how we first reported it. It is not one. Both statements are instances of a single quantity — the share of the estimand's own support that the outage hides — and the apparent dichotomy is what that quantity looks like when you evaluate it at a 30-day support and at a 365-day support and nowhere in between.

The insight of this paper is one sentence: a record-screening rule should condition on the share of the target statistic's support that the outage hides, and because that support is a property of the statistic rather than of the record, the governing axis moves continuously with the statistic's form. Figure 1 states the argument; Figure 5a measures it. Sweeping the hottest-k-day mean from $k = 1$ to $k = 365$ traces the whole scale, and the crossover between the two regimes is a number we can report: 91 days.

Two consequences run through the rest of the paper. The first is practical: a support has to be *located* before it can be conditioned on, and a station's hottest window moves between years. Conditioning on the estimand's own support is therefore not optimal for a narrow estimand; conditioning on that support dilated by the year-to-year wander of its location is, and one 60-to-90-day window serves every endpoint we examined. The second is that screening is not the only lever available, and on our panel it is not the largest one: when real outages happen turns out to matter about as much as which of them are screened out.

Existing work cannot settle this, for structural rather than incidental reasons. Guidance measures tolerance along one axis, so it cannot compare axes. Gap-filling studies optimise the accuracy of the interpolated series rather than the admissibility of the completed year (Mohseni et al., 1998; Li et al., 2017). Beaufort et al. (2022) come closest: they gap-fill incomplete summer records with air-temperature regressions precisely so that a thermal peak can be computed, and they use the same statistic we do — the mean of the thirty hottest consecutive days — validating the filling on fully monitored reference sites. But their object is spatial extrapolation

across 105,800 reaches; the filling is a fixed preprocessing treatment applied to every record alike, and no station-year is ever declined. Large-sample thermal databases (Isaak et al., 2017) and the machine-learning stream-temperature literature — 57 studies between 2020 and 2024 by the count of Corona and Hogue (2025) — optimise prediction skill at daily or reach scale and rarely propagate to a derived annual statistic, let alone to an admission decision. What is common to all of them is that the screening decision is never conditioned on what the target statistic actually depends on.

Our contributions are at five levels.

1. **A law with a measured parameter.** The share of the estimand's support that an outage hides governs the error of the reported statistic; the familiar timing-versus-length contrast is its behaviour at the two ends of a scale, and the crossover sits at a support width of 91 days.
2. **A correction to the law that makes it deployable.** The support must be dilated by the year-to-year wander of its own location, which we measure; one 60-to-90-day climatological window is near-optimal for every endpoint from the annual maximum to the half-year mean.
3. **An explanation of the standing guidance's own anomaly, on its own estimands.** Reproducing Johnson et al.'s sine-fit mean, phase and amplitude on our panel, the three have three different supports, and a timing effect appears for one and not the others — which is what "inconsistent in space and time" looks like from inside.
4. **A rule taken to the records it is for.** Pre-registered, scored once on 36 reserved networks, and then applied to 2,422 genuinely incomplete station-years that carry no truth and have never been decided before.
5. **Two boundaries stated as prominently as the result.** The operational gain is regime-dependent and is not resolvable under the outage geometry these networks currently experience, and the reason is a finding in itself; and a screen that selects on difficulty selects against mountain catchments unless it is stratified, which we price.

2. Related work: a map of what is being optimised

The literature this paper intervenes in is not one conversation but five, and they differ in what they optimise, on what unit they decide, and along which axis they condition.

body of work	optimises	decision unit	conditioning axis	reports uncertainty on the derived metric
record-completeness guidance (Johnson et al., 2021)	tolerable amount of missing data	station-year	outage size	no
agency collection standards (Mauger et al., 2015)	comparability across a monitoring programme	deployment	season and duration	no

body of work	optimises	decision unit	conditioning axis	reports uncertainty on the derived metric
gap filling and record extension (Mohseni et al., 1998; Li et al., 2017)	accuracy of the filled series	day	air temperature, donors	rarely
thermal-peak reconstruction and extrapolation (Beaufort et al., 2022)	spatial coverage of the thermal peak	reach	air-temperature regression, applied uniformly	validation against reference sites, not per case
large-sample thermal databases and models (Isaak et al., 2017; Corona and Hogue, 2025)	prediction skill	day or reach	meteorology, catchment attributes	occasionally, at daily scale
reject-option and selective prediction (Geifman and El-Yaniv, 2017; Zaoui et al., 2020)	error rate at a fixed rejection rate	case	model confidence	as a rejection rate, not in physical units
this work	**error of the derived metric among retained station-years**	**station-year**	**the share of the estimand's support the outage hides**	**yes, as a calibrated interval in °C**

Four consequences follow from the map. First, no existing line compares screening axes, because each fixes one. Second, uncertainty on the *derived* statistic — the number a manager reads — is essentially unreported; daily skill is reported instead, and the two are not interchangeable for an extreme. Third, the selective-prediction literature supplies decision machinery whose control variable is a rejection rate, whereas an analyst needs a tolerance in degrees on the quantity they report. Fourth, and least often said, the missing-data literature already contains the object we need: the influence function of an estimator (Hampel, 1974) says exactly how much each observation moves the estimate, and whether missingness is ignorable depends on the mechanism that produced it (Rubin, 1976). What has not been done is to use either as a screening rule for archived environmental records. This paper occupies the row that is otherwise empty.

3. Problem formulation

Let a station-year be a sequence $y = (y_1, \dots, y_T)$ of daily mean water temperatures, $T \in \{365, 366\}$, and let $q(\cdot)$ be the derived statistic of interest — here the hottest consecutive k-day mean, $q_k(y) = \max_t \frac{1}{k} \sum_{i=t}^{t+k-1} y_i$, which is the annual maximum at $k = 1$ and the annual mean at $k = T$, together with the sine-fit mean, phase and amplitude as alternatives.

An outage is a binary mask $m \in \{0, 1\}^T$; the analyst observes $y_{\bar{m}}$, the days where

$m_t = 0$. A *reconstruction* $\hat{y} = R(y_{\bar{m}}, Z)$ completes the year using side information Z — synchronous donor stations, air temperature, the station's own history — and the analyst reports $q(\hat{y})$. The quantity that matters is the error of the reported statistic,

$$L = \big|q(\hat{y}) - q(y)\big|,$$

and a station-year is *consequential* when $L > \tau$ for a tolerance τ stated in the units of q.

A **screening rule** is a map $s : (m, Z) \mapsto \{\text{retain}, \text{exclude}\}$ that must not read the hidden days y_m. Any such rule traces a retention–error frontier: at retention $r = \mathbb{E}[s = \text{retain}]$, the quantity of interest is $\Pr(L > \tau \mid s = \text{retain})$. Two rules are compared at equal r, because a rule can always buy accuracy by discarding more.

The incumbent is $s_{\text{len}}(m) = \mathbb{1}\{\max \text{run}(m) \le \theta\}$, a threshold on outage size. Its conditioning set contains no information about *where* the outage fell relative to the part of the record q depends on.

3.1 The support of an estimand, and the hidden mass

Write $w_t^q = \big|\partial q(y)/\partial y_t\big|$ for how much day t moves the statistic — the discrete influence function of q at y. The **hidden mass** of an outage is the share of that influence the outage removes,

$$h_q(m) \;=\; \frac{\sum_{t : m_t = 1} w_t^q}{\sum_t w_t^q}.$$

For the annual mean every w_t is equal and h_q is the missing fraction. For the hottest k-day mean the influence is concentrated on the k days that achieve the maximum, and h_q is the share of that window the outage hides. For the sine-fit parameters the weights are smooth functions of day of year: a constant for the fitted mean, $|\sin(\theta_t - \phi)|$ for the amplitude, which is largest at both extremes, and $|\cos(\theta_t - \phi)|$ for the phase, which is largest a quarter-year from the peak. The central hypothesis is about this quantity.

> **H1.** $\Pr(L > \tau)$ is governed by h_q. As the support of q widens from a single day to the whole year, the discriminating power of a fixed narrow window decays, crosses that of a whole-year completeness measure, and ends below it.

H1 is falsifiable in several places at once: it fails if a fixed peak window discriminates equally well at every support width, if the crossing does not occur, if the ordering is not stable in τ, or if h_q computed from the influence function does not beat both hand-chosen axes where the influence is not uniform.

3.2 Estimated supports, and why the right conditioning set is wider

h_q as written is not available at decision time, because the influence weights depend on y, including the days the outage hides. What an analyst has is the *climatological*

support: the window the station's other complete years say the maximum should fall in. That is an estimate, and it has an error. Writing $\hat{W}_q$ for the estimated support and σ for the year-to-year wander of its location, conditioning on $\hat{W}_q$ alone underweights days that the true support may have reached. The correction is to dilate: condition on $\hat{W}_q$ widened by a multiple of σ. §5.3 measures σ and the dilation that follows from it, and the result is that a single window serves the whole family.

4. Data and study design

Figure 3 gives the design as a whole.

4.1 Records and quality control

Daily water-temperature series were assembled from nine national providers: ARSO (Slovenia), CHMI (Czechia), FOEN (Switzerland), GKD Bayern and LUBW (Germany), NVE (Norway), Rijkswaterstaat (Netherlands), SYKE (Finland) and the USGS (United States). A *network* is one provider's group of stations on a common river system; it is the unit at which donor stations are available to one another, and it is the clustering unit for every interval reported here, because station-years within a network share weather, regulation and instrumentation.

Waterbody class was assigned per network and non-river objects — seas, lakes, canals, estuaries, deltas — are excluded rather than averaged into a river result. Reference series are never imputed: a calendar year with any missing day supplies no truth. Stations with fewer than four fully observed calendar years are excluded because the estimator protocol requires them. The full denominator with every exclusion and its count is in the supporting information.

This leaves **146 networks, 457 stations and 5,114 fully observed station-years**. The same 457 stations contribute **2,422 further station-years that are genuinely incomplete**. Those carry no truth and are therefore not part of the scored population; §5.8 decides them anyway, because they are the population a screening rule exists for.

Station coordinates were assembled for all 457 stations — from the providers' own catalogues where the repository held them, from the USGS NWIS site service for 195 United States gauges, and from ARSO's public station feed — and daily 2 m air temperature was joined from NASA POWER for the whole panel, 1981 to 2026. The v17 programme had air temperature for 2.3% of development cases; it now has it for **98.6%**, which matters because air-temperature regression is the reconstruction this field actually uses (§5.6).

For the United States subset, basin characteristics were joined from GAGES-II (Falcone, 2011) on the eight-digit gauge identifier, matching **140 of 204** eligible stations across 57 networks and supplying baseflow index, subsurface contact time, drainage area, site elevation, basin slope and upstream dam storage. These are used in §5.4 and §5.9 to ask which rivers are hard and which rivers a screen keeps; they are deliberately *not* used inside the screening rule, because they are unavailable for the other eight agencies.

4.2 Split

Networks — never station-years — were split into development and reserved, stratified on provider and on donor regime (no synchronous donors, one to two, three or more), with the seed recorded before any scoring. This gives **110 development and 36 reserved networks** with all three donor regimes present in both. The reserved panel was opened once, after the analysis was frozen, and the model was not retuned. Everything added after that opening — the support sweep, the sine endpoints, the archive deployment, the representativeness analysis, the air forcing — is reported on development cases, with reserved numbers shown only where they are informative and labelled as no longer blind.

4.3 Endpoints and their natural scale

The primary endpoint is the hottest consecutive 30-day mean. For the support sweep it is generalised to the hottest consecutive k-day mean for $k \in \{1, 3, 7, 15, 30, 60, 90, 182, 365\}$; the coldest 30-day mean, the annual amplitude, and the three sine-fit parameters are scored alongside.

A tolerance in degrees means nothing without a scale, so each endpoint is scored against a tolerance equal to the same fraction — 0.47 — of *its own* interannual standard deviation at the same stations. For the annual peak that scale is **1.06 °C**, giving a primary tolerance of **0.50 °C**; the scale falls monotonically with support width, from 1.18 °C at $k = 1$ to 0.55 °C at $k = 365$. The fraction is swept at 0.25, 0.47 and 1.0 and no conclusion about which axis wins is claimed unless it is invariant across the sweep.

The reference series carries instrument error of roughly 0.1–0.2 °C, so the primary tolerance is about three times sensor accuracy; §5.10 perturbs the reference by 0.15 °C and repeats the analysis.

4.4 Outage design

A case is one station-year under one outage realization. Two arms are used and reported separately, because they are not one population. The **empirical** arm samples real outage runs — length and start month — from the same network's own records and transplants them onto fully observed years. The **stratified** arm crosses outage length {14, 30, 60, 90, 120, 180, 270 days} with the overlap between the outage and the station's climatological peak window, so that the response surface is estimable rather than accidental. The design yields **51,524 cases**, 40,704 development and 10,820 reserved.

Both arms transplant a mask onto a year whose values were observed, which assumes the mask is uninformative about the values it hides. §5.10 tests that assumption against the archive rather than asserting it. Outages are placed inside one calendar year; failures that straddle New Year damage two annual statistics while hiding neither warm season, and the design cannot represent that case.

4.5 Estimators

Every case is completed by each supported estimator: the statistic computed from recorded days alone (current practice), a day-of-year median climatology, a smoothed climatology, linear interpolation where both flanks exist, ridge regression on the visible subset of synchronous donors, and an air-driven ridge with lagged air temperature and harmonics. Estimators are fit leave-one-year-out within the station, which is what offline archive repair permits; §5.10 repeats the study under the stricter protocol in which only prior years are available. Support is declared per method: donor regression does not read a flanking observation, so requiring one would discard usable cases, and absent donor columns are never imputed but the model is refit on the visible subset.

4.6 The screening rule

Figure 2 is the architecture. For each case the screen predicts an upper bound on the absolute error each supported estimator would produce, selects the estimator with the smallest bound, and emits a conformal interval on the statistic together with an admission decision at an analyst-chosen tolerance. Its inputs are outage geometry, hidden mass on the estimated support, warm-season coverage, the margin by which the observed days already constrain the annual maximum, donor count and coverage and training-period deseasonalised correlation, and station history. No input may read a day the outage hides, and that restriction is enforced in code.

Calibration is split conformal with two properties that matter. It is *selection-conditional* — the bound is calibrated after the estimator is chosen, because taking the smallest of several bounds is a winner's curse that per-estimator calibration cannot see — and *multiplicative*, because the error scale spans an order of magnitude between an outage that misses the peak window and one that covers it. Fitting, calibration and test networks are always disjoint.

4.7 Action-space screening before rule fitting

Before any rule was fit we measured, on development cases, how much a rule with perfect foresight could win inside each candidate action space, and compared it with the margin the reserved panel can resolve. Two natural framings of this problem — adopt or reject a fixed reconstruction, and choose among reconstruction methods — have ceilings of +0.038 °C and +0.080 °C against a threshold of 0.136 °C, and were not pursued. Retain-or-exclude clears its gate by a factor of five and is the action space studied here. We recommend the check as standard practice: a decision experiment whose ceiling is below its own resolvable margin cannot produce evidence in either direction, and running one consumes a reserved panel for nothing.

4.8 Inference

Every interval resamples whole networks, because station-years within a network are not independent and reusing a station-year under several outage realizations does not create independent observations. The pre-registered primary criterion, frozen before the reserved panel was opened, is that at the retention of the matched length rule the screen must reduce the consequential-error rate by at least five

percentage points with a network-clustered interval excluding zero. Precision was computed in advance: the reserved panel resolves 1.44 points, so the declared effect is 3.5 times the resolution.

Where several endpoints are compared, the direction of each contrast was predicted in advance from the form of the statistic, which is the defence against multiplicity; the comparisons are confirmatory of a stated prediction rather than a search for the favourable cell. Three of the predictions declared for this round were not met, and §5.3 and §5.5 report them as failures rather than reshaping them into successes.

5. Results

Results are organised by research question. RQ0 asks whether the problem exists, RQ1 tests the law and RQ1b tests it on the standing guidance's own estimands, RQ2 asks which rivers are hard, RQ3 to RQ5 evaluate acting on the law, RQ6 takes the rule to records that really have holes in them, and RQ7 to RQ9 establish where it fails.

5.1 RQ0 — Does the problem exist, and how large is it?

Before proposing any rule, we measure what current practice costs. On development cases, computing the annual peak from recorded days alone — the default once a station-year passes a completeness screen — gives a mean absolute error of **1.305 °C, or 1.23 interannual standard deviations** of the quantity being measured, with **24.9%** of station-years wrong by more than 1.0 °C. Substituting a day-of-year climatology over the outage gives 0.402 °C; the best supported reconstruction gives 0.225 °C once air forcing is available for the whole panel (0.242 °C without it); an oracle over the estimator set gives 0.130 °C.

The problem is therefore real and large, and its largest single component is not any decision rule: it is that thermal statistics are routinely computed on records with holes in them. Everything that follows concerns the smaller, second-order question of which of the remaining station-years to trust.

5.2 RQ1 — What governs the error, and does it move with the estimand?

The dichotomy first. Table 1 reports, for five endpoints, how well each candidate axis discriminates a consequential error, with the contrast between the best timing axis and outage length carried by a network-clustered bootstrap.

Table 1. Discrimination of a consequential error by axis, development cases, each endpoint at 0.47 of its own interannual standard deviation.

endpoint	peak-window overlap	warm-season coverage	outage length	AUC(timing) − AUC(length), 95% CI	governing axis
hottest 7-day mean	0.823	0.824	0.752	**+0.072 (0.063, 0.096)**	timing
hottest 30-day mean	0.819	0.820	0.750	**+0.070 (0.061, 0.098)**	timing

endpoint	peak-window overlap	warm-season coverage	outage length	AUC(timing) − AUC(length), 95% CI	governing axis
amplitude	0.758	0.792	0.768	**+0.024 (0.015, 0.037)**	timing
annual mean	0.672	0.818	**0.845**	**−0.027 (−0.036, −0.018)**	length
coldest 30-day mean	**0.447**	0.619	**0.803**	**−0.184 (−0.228, −0.156)**	length

All five intervals exclude zero and the ordering reverses, as the support account predicts. The ordering is also invariant to where the tolerance is drawn: at 0.25, 0.47 and 1.0 interannual standard deviations the winning axis is unchanged for every endpoint.

Two features deserve emphasis. The **seven-day peak has the largest timing advantage of any endpoint**, which the mechanism predicted before it was measured: a shorter extreme is more localised in time. And **peak-window overlap has an AUC of 0.447 for the cold extreme** — hiding the summer peak carries no useful information about the winter minimum, exactly as it should not.

Now the continuum. Table 1 evaluates the law at two points and reads the gap between them as a dichotomy. Sweeping the hottest-k-day mean from $k = 1$ to $k = 365$ evaluates it everywhere (Figure 5a).

Table 2. The governing axis as a function of the support of the estimand, development cases.

support width k (days)	interannual SD (°C)	AUC, 30-day peak window	AUC, whole-year missing fraction	difference, 95% CI
1	1.18	0.825	0.751	**+0.074 (0.053, 0.097)**
3	1.17	0.824	0.746	**+0.078 (0.055, 0.104)**
7	1.16	0.823	0.746	**+0.077 (0.053, 0.104)**
15	1.12	0.821	0.743	**+0.078 (0.053, 0.106)**
30	1.08	0.819	0.744	**+0.075 (0.049, 0.104)**
60	1.02	0.815	0.765	**+0.050 (0.028, 0.078)**
90	0.94	0.792	0.790	+0.002 (−0.018, 0.027)
182	0.74	0.728	0.823	**−0.095 (−0.111, −0.077)**
365	0.55	0.672	0.845	**−0.173 (−0.192, −0.157)**

The advantage of the fixed peak window is flat at about +0.075 AUC for every es-

timand whose support is a month or less, decays through +0.050 at a two-month support, and **crosses zero at a support width of 91 days**; beyond that the whole-year completeness axis wins, by 0.095 AUC at a half-year support and 0.173 at the annual mean. Every interval excludes zero except the one at $k = 90$, which is where the crossing is. There is no dichotomy to explain. There is one scale with a measured crossover, and the two regimes the previous account named are its ends.

Is the mechanism a measurement? Binning development cases on both covariates simultaneously and measuring the growth of error along each axis with the other averaged out:

endpoint	error ratio, most days missing vs fewest	error ratio, peak window hidden vs untouched
hottest 30-day mean	3.5×	**9.6×**
annual mean	**10.2×**	1.5×

The two ratios swap by roughly an order of magnitude in each direction. For the peak, once timing is in the model, outage length adds nothing measurable: its permutation importance is **−0.004** against 0.194 for peak overlap (Figure 4c), and nested feature sets grouped by network rise from AUC 0.742 with length alone to 0.845 adding timing, 0.858 adding the observed-peak margin and 0.884 adding donor quality (Figure 4b). Two qualifications belong with that. Length is not irrelevant: holding peak overlap at zero, the consequential-error rate still rises from 0.6% at 1–30 days to 8.7% at 121–200 days. And in a linear score that already contains warm-season coverage the length term takes a small negative weight, which is collinearity — the two correlate at −0.86 — and not evidence that longer outages are safer. That correlation is also why the previous labelling was wrong: warm-season coverage is a *count of missing days restricted to a window*, which is a length measure with a domain restriction, and it is the domain restriction rather than the timing that does the work.

Donor information enters through quality rather than quantity. Halving the donor set by training-period correlation while holding the count fixed costs **+0.082 °C** (95% interval 0.061–0.112). Replacing donor anomalies with phase-shuffled surrogates while preserving each donor's seasonal cycle costs **+0.234 °C** (0.164–0.288), so donors supply synchronous anomaly information and not merely a seasonal shape (Figure 11b).

5.3 RQ1b — Two predictions that failed, and what they buy

Two of the predictions declared for this round were not met. Both failures are informative and both are reported here rather than in a supplement.

A narrow estimand is best screened on a window wider than itself. We predicted that the axis restricted to the estimand's own support would be at least as good as any hand-chosen axis at every k. It is not: for supports of 15 days or less the fixed 30-day peak window beats the estimand's own support, by 0.064 AUC at $k = 1$. Table 3 gives the whole matrix of estimands against conditioning windows and the reason.

Table 3. Which climatological window a screen should condition on, development cases. The diagonal is the estimand's own support.

estimand, hottest k-day mean	best conditioning window (days)	AUC there	AUC on its own support	gain from widening	year-to-year wander of the support's location (days)
1	90	0.862	0.760	+0.102	16.7
3	60	0.858	0.766	+0.092	16.3
7	90	0.855	0.778	+0.077	16.5
15	60	0.853	0.802	+0.051	15.3
30	90	0.852	0.827	+0.025	13.7
60	90	0.860	0.855	+0.005	9.4
90	90	0.856	0.856	0	7.4
182	182	0.851	0.851	0	6.9
365	365	0.844	0.844	0	—

The gain from widening tracks the wander almost exactly: it is largest where the climatological window is the worst estimate of where this year's window will fall, and vanishes once the support is wide enough to contain its own location error. A screening rule does not condition on the support; it conditions on the support *dilated by the uncertainty in the support's location*, and a dilation of roughly two location standard deviations on each side is what the data choose. One practical consequence stands out: **AUC at the best conditioning window is between 0.85 and 0.86 for every estimand from the annual maximum to the half-year mean.** An agency needs one climatological window of 60 to 90 days, not one per metric. It also means the screen deployed in §5.6, which conditions on a 30-day window, is leaving 0.025 AUC on the table for its own endpoint.

A hand-chosen transition window is not enough for a phase. We predicted that for the sine-fit phase — whose influence is largest a quarter-year from the peak — two 61-day windows around the spring and autumn limbs would beat both peak overlap and outage length. They beat peak overlap decisively (+0.249 AUC, 95% CI 0.231 to 0.270) and they do not beat outage length (0.831 against 0.836). The phase's influence is spread over half the year in two broad lobes, so a hand-drawn window discards too much of it. Weighting every day by the influence function itself recovers the advantage: 0.847, which beats the whole-year missing fraction by +0.013 (0.009, 0.017) and peak overlap by +0.265 (0.249, 0.285). The lesson is that "condition on the support" is exact and "condition on a window near the support" is an approximation whose quality depends on how concentrated the influence is.

5.4 RQ1c — The standing guidance's own estimands

Johnson et al. (2021) fit a sine wave by linear regression and take a mean, a phase and an amplitude. Reproducing that estimator on our panel puts our claim and their guidance on the same footing for the first time (Figure 5c).

Table 4. Axis comparison on the sine-fit parameters, development cases, each at 0.47 of its own interannual standard deviation.

estimand	30-day peak window	whole-year missing fraction	outage length	influence-weighted hidden mass	best axis
fitted mean	0.672	0.845	**0.845**	0.845	outage length
amplitude	0.724	0.812	0.815	**0.831**	influence-weighted
phase	0.583	0.835	0.836	**0.847**	influence-weighted

Three things follow, and the first is the most important for how this paper should be read. **On the estimands the guidance was derived for, the guidance is right.** A completeness threshold is the correct axis for the fitted mean and is within 0.02 AUC of the best axis for the phase. Our earlier framing — that the incumbent rule "gets an ordering exactly backwards" — is true only for warm-season extremes, which Johnson et al. never studied. What we are correcting is the transfer of a threshold across estimands, not the threshold.

Second, the ordering is the same under the authors' own protocol, in which the sine is fitted to whatever days are present with no repair at all: peak overlap 0.667, 0.743 and 0.590 against whole-year missingness 0.902, 0.842 and 0.898 for the mean, amplitude and phase. The base rate is far higher — 32% to 34% of station-years consequential, against 10% to 12% after repair — which is a separate argument for repairing before screening.

Third, this explains the anomaly the guidance reported. Johnson et al. observed that the timing of missing data mattered "inconsistently in space and time". They were observing three parameters at once whose supports are a constant, a pair of lobes at both extremes, and a pair of lobes at the transitions. A timing effect appears for the amplitude and not for the mean, and a peak-window effect appears for none of them. Inconsistency is what a mixture of three different supports looks like from inside. We do not contradict the observation; we decompose it, and we can now say in advance which of their parameters should show a timing effect.

5.5 RQ2 — Which rivers are hard?

The mechanism above is a property of estimators and estimands and would hold for any time series. This section asks the hydrological question. For the 140 United States stations matched to GAGES-II across 57 networks, we regressed donor synchrony and completed-year error on basin attributes, with the expected direction of each effect declared before the regressions were run (Figure 13).

Topography, not geology, predicts difficulty. Site elevation and basin slope both reduce donor synchrony (Spearman $\rho = \mathbf{-0.29}$, 95% CI −0.46 to −0.08, and $\rho = \mathbf{-0.31}$, −0.51 to −0.07), and site elevation raises both completed-year error ($\rho = +0.24$, 0.06 to 0.43) and the consequential-error rate ($\rho = +0.25$, 0.08 to 0.43). Mountain stations are harder because their neighbours are less exchangeable: steeper lapse rates, snowmelt timing, aspect and shading make two nearby gauges less alike.

The groundwater prediction from the literature is not reproduced, and the reason is instructive. Johnson et al. report that filling can harm groundwater-influenced sites, which predicts that a high baseflow index implies larger completed-year error. Baseflow index does reduce donor synchrony in the predicted direction, from 0.70 in the lowest tertile to 0.59 in the highest, although the rank correlation interval includes zero ($\rho = -0.19$, -0.42 to 0.10). But the error moves the *other* way: high-baseflow stations are **−0.043 °C** easier than low-baseflow stations (95% CI −0.081 to −0.004).

Two opposing effects explain it, and the second is larger. Groundwater contribution makes donors less informative, but it also damps the thermal regime itself: across baseflow tertiles the mean annual peak falls from 22.7 °C to 18.1 °C ($\rho = \mathbf{-0.47}$, -0.60 to -0.26) and its interannual standard deviation from 0.98 °C to 0.76 °C. A damped regime is a smaller and steadier target, and that outweighs the loss of donor information (Figure 13c). We therefore do not contradict the earlier observation so much as decompose it: groundwater influence degrades the *information* available for reconstruction while simultaneously reducing the *difficulty* of the quantity being reconstructed.

Regulation does not make station-years harder for this endpoint: reference and non-reference basins differ by +0.004 °C (95% CI −0.051 to +0.093).

Catchment attributes explain a modest share of the variation — a standardised linear model reaches $R^2 = 0.22$ for donor synchrony and 0.13 for completed-year error — and **no attribute predicts how much the screen gains**: every correlation between a basin attribute and the per-station screening benefit has an interval containing zero. The screening problem is therefore governed mainly by outage geometry rather than by catchment physics, which is why basin attributes are reported here as an explanation and are excluded from the screening rule. They return in §5.9, where the question is not what the screen gains but whom it keeps.

5.6 RQ3 — Does screening on the governing axis pay?

At 70% retention on development cases, the share of retained station-years with a peak error above 0.5 °C is 2.9% for the calibrated screen, 5.6% for a rule on peak overlap alone, 9.2% for the outage-length rule and 15.0% for random retention (Figure 6a). On the **36 reserved networks, opened once with the frozen rule and no retuning**, the screen gives 2.5% against 9.7% for the length rule (Figure 6b): a contrast of **−7.13 points, 95% network-clustered interval −8.97 to −5.36**, against a pre-registered effect of five points and a resolvable margin of 1.44 points. Both pre-registered conditions are met, in every donor regime (none −6.6, sparse −6.7, rich −7.2, all excluding zero).

Placed at the operating points the published guidance actually implies, and not at an arbitrary retention: at the seven-week unfilled limit (49 days, 49.2% retention on the reserved panel) the length rule leaves 5.2% and the screen 0.3%; at the twenty-week imputed limit (140 days, 85.1% retention) the length rule leaves 11.3% and the screen 6.9%. The second of those numbers is new to this version and it matters: the v17 manuscript placed this comparison at 91 days, having mis-stated the imputed limit as thirteen weeks rather than twenty. The corrected limit is far

more permissive — it admits 85% of cases rather than 74% — and the screen's advantage there is 4.4 points rather than 6.2.

Table 5. Reserved panel at 70% retention, across tolerances.

screen	> 0.25 °C	> 0.5 °C	> 1.0 °C	underestimate > 1 °C
outage-length rule	17.2%	9.7%	3.4%	**2.0%**
peak-overlap rule	9.2%	4.2%	1.2%	0.7%
calibrated screen	7.1%	2.5%	0.4%	**0.3%**
random	23.9%	14.4%	5.6%	3.7%

Understating a summer peak by more than a degree converts a thermally stressful year into an unremarkable one. The screen reduces that rate sevenfold relative to the incumbent.

The panel is unbalanced — one provider contributes 56 of the 110 development networks — so the contrast was recomputed with each unit weighted equally. On the reserved panel: case-equal −7.01, provider-equal −6.29, network-equal −5.82, station-equal −5.54 points. Every weighting clears the declared five points and every interval excludes zero. Across individual reserved networks the median contrast is −5.0 points, **83% of networks favour the screen**, and the worst is +2.8.

5.7 RQ3b — Was the incumbent handicapped, including by the absence of the field's own method?

A screening comparison can be rigged by pairing the incumbent with a weak reconstruction, and the v17 estimator library had a specific weakness: air temperature was available for 2.3% of development cases, so the reconstruction this field actually uses was effectively absent. With coordinates joined for all 457 stations and NASA POWER pulled for the whole panel, air forcing now covers **98.6%** of development cases and the comparison can be rerun.

Table 6. Development cases at 70% retention, with air forcing available throughout.

screen	reconstruction	error rate above 0.5 °C
length rule	information-conditioned policy (used throughout)	9.00%
length rule	**always air regression (Beaufort-style), else climatology**	**15.16%**
length rule	always donor ridge, else climatology	9.57%
length rule	always smooth climatology	19.79%
length rule	**oracle over methods, not deployable**	**3.62%**
calibrated screen	deployable estimators it selects itself	**2.87%**

The screen beats the incumbent armed with a reconstruction nobody can run, and the reconstruction paired with the incumbent throughout is the strongest of the

deployable options rather than the weakest. Pairing the incumbent with a Beaufort-style air regression applied to every record alike makes it *worse*, not better, which is the same point from the other side: a reconstruction that is right on average is not right per case.

Three further observations belong here. First, the axis conclusion does not depend on the estimator library — the timing-minus-length AUC contrast is +0.070 with the old forcing and +0.081 with the new. It is a claim about which cases are hard, not about how well they are repaired. Second, the screen uses the new information: it selects the air-driven estimator for **20.0%** of cases, against 0.07% in v17, and donor regression for 53.1%. Third, and least comfortable, the *fixed* information-conditioned policy gets slightly worse at donor-less stations when air becomes available (0.340 °C to 0.364 °C), because its preference order was written when air was rare and it now prefers an air regression to a climatology at sites where the climatology is better. A fixed ladder over estimators is not robust to a change in what is available; a per-case selection is. That is an argument for the method of this paper rather than against it, and we report it because it was not the result we expected.

The pre-registered confirmation is **not** re-scored against the new forcing. It stands as reported in §5.6; this section is a development-panel sensitivity analysis.

5.8 RQ4 — What does each component contribute?

Table 7. Component ablation, development cases at 70% retention, each variant re-run through the full out-of-fold pipeline.

screen	error rate	points vs the length rule
outage-length rule (incumbent)	9.24%	—
three-variable logistic regression	5.53%	−3.71
peak-overlap alone, no fitting at all	5.62%	−3.62
three-feature gradient boosting	4.89%	−4.35
geometry and timing only (8 features)	4.86%	−4.38
drop donor features	3.78%	−5.46
drop the observed-peak margin	3.13%	−6.11
full screen	2.93%	−6.31
full screen, screening only (no method choice)	3.11%	−6.13

Fifty-nine per cent of the benefit is available from a logistic regression on three variables, and 57% from peak-window overlap with no fitting at all. Method selection contributes 0.2 points. The contribution of this paper is therefore the conditioning set, and the calibrated machinery is a refinement on it; we report the simple rule as a deliverable rather than as an ablation. Fitted on development networks and applied unchanged to the reserved panel,

```
risk = logistic( −1.302 + 2.070 · peak_overlap
                        − 2.412 · warm_observed_fraction
                        − 0.00187 · longest_run_days )
```

gives 4.2% against the length rule's 9.6% at 70% retention, a contrast of **−5.44 points (−6.85, −3.96)**, and cuts severe underestimation of the peak from 1.9% to

0.6%. Three quantities an analyst can compute in a spreadsheet clear the effect size this study pre-registered for its full method.

One interpretable piece of the remainder: the screen selects linear interpolation for 25.0% of reserved cases against 5.7% under a presence-conditioned rule, and interpolation carries the lowest error of any choice when selected (0.10 °C). For short outages with both flanks observed, interpolation beats donor regression.

5.9 RQ5 — Where is the advantage absent, and why?

The pre-registered test was run on the pooled case set, which mixes two outage arms whose base rates differ by an order of magnitude. They must be reported apart.

Table 8. Decomposition by outage arm.

panel	arm	cases	base rate, retain all	length rule at 70%	screen at 70%	contrast (95% CI)
development	empirical	14,995	7.4%	2.52%	0.38%	−2.13 (−3.45, −1.26)
development	stratified	25,709	19.3%	14.58%	6.17%	−8.39 (−9.25, −7.47)
reserved	empirical	4,041	4.2%	0.74%	0.11%	**−0.71 (−2.36, +0.05)**
reserved	stratified	6,779	20.8%	16.59%	7.78%	−8.68 (−10.46, −6.51)

The declared five-point effect is met on the pooled and stratified populations and **not** on the empirical arm, where the reserved interval includes zero. The headline −7.1 points is therefore an effect on a mixture that includes a designed arm, and it is not an estimate of what any agency will experience next year. The mechanism is general; the operational gain is regime-dependent.

Why the empirical arm is null is itself a result. The v17 manuscript reported the null and left it unexplained. Across the 6,472 real interior gaps in this archive, gaps shorter than 30 days and gaps of 30 to 89 days are close to uniform in start month ($\chi^2 = 4.0$ on 11 df, $p = 0.97$ for the 30–89 day class). Gaps of 90 days or more are not: **72.3% of them begin between September and January and 17.9% between March and July** ($\chi^2 = 86.1$, $p = 9.5 \times 10^{-14}$). The outages long enough to destroy a summer peak are seasonal shutdowns, and they start after the warm season has been recorded. Agencies are already protecting the summer peak, by a calendar nobody chose for that reason. Under that calendar little is at risk, so little can be screened out: 4.2% of reserved empirical cases exceed the tolerance with no screening at all.

Figure 7 shows that calendar and what it is worth.

The screen earns its place in the regime that motivates screening — long outages overlapping the warm season — and §5.10 shows how much of a real archive falls in it.

5.10 RQ6 — What the rule does to records that really have holes

Every number above is computed on a complete station-year with a synthetic hole punched in it, which is the only way to have a truth. The same archive contains **2,422 genuinely incomplete station-years**, a third of the total; they carry no truth and have never been decided. They are the population a screening rule exists for, and three things can be asked of them without inventing a truth (Figure 8).

They are not a marginal population. The median incomplete year is missing 9 days and the 90th percentile 127. **28.7%** of them have at least one day of the station's climatological peak window missing, and 261 have more than half of it missing.

The two rules disagree on a sixth of them. At the seven-week unfilled limit the length rule admits 83.2% and a peak-window rule at 10% hidden admits 81.2% — near-equal retention, which is the comparison this paper insists on — and they **disagree on 15.1%** of station-years: **207 are admitted by the incumbent and rejected on timing**, and 158 are rejected by the incumbent and admitted on timing. At the twenty-week imputed limit the incumbent admits 91.7% and the disagreement is 11.6%, of which 267 are in the dangerous direction and 13 in the wasteful one. The disagreement is present in every provider with more than 20 incomplete station-years, from 11.1% (GKD Bayern) to 25.7% (SYKE).

Table 9. Archive composition under each admission rule, on the 2,422 real incomplete station-years.

rule	admitted	admitted with any peak-window day hidden	with more than half hidden	with the peak window wholly hidden
repair everything, no screening	2,422	696	261	**186**
length ≤ 140 d (published, with imputation)	2,221	507	82	**17**
length ≤ 49 d (published, unfilled)	2,016	437	45	2
peak window ≤ 10% hidden	1,967	241	**0**	**0**

The last two columns are the practical case for the paper, and they need no truth. The published guideline that agencies are most likely to apply — the twenty-week limit, which exists because imputation is assumed — admits 17 station-years whose annual peak was never observed at all and 82 whose peak window is mostly gone. Each of those contributes an annual peak to an archive that was inferred rather than measured, and nothing in the record marks it.

Transporting the risk model. A model of P(consequential error) fitted on development cases from decision-time features only, and checked for calibration on the reserved panel first (largest decile miscalibration 6.2 points, and under 1.6 points in

nine of ten deciles), predicts a consequential-error rate of **0.98%** among the years the unfilled limit admits, **1.85%** among those the imputed limit admits, and **0.26%** among those the peak-window rule admits at essentially the same retention. This is an extrapolation from scored cases to unscored records and is labelled as one; §5.12 tests the exchangeability it assumes.

What changes in the regional picture is small. Admitting incomplete years at all raises the archive's mean annual peak by about half a degree relative to the complete-year subset, and the choice of rule moves that by less than 0.1 °C: +0.52 °C repairing everything, +0.51 °C under the unfilled limit, +0.53 °C under the imputed limit and +0.45 °C under the peak-window rule. The share of station-years above 20 °C moves from +4.55 points to +4.17 points across the same four rules. We report this because it deflates a claim we might have made: at the level of a regional aggregate, the decision that matters is whether to repair, not which rule screens afterwards. The rule matters per station-year, and it matters for who is in the archive (§5.11), not for the mean.

5.11 RQ7 — What changes downstream

Per-year judgements. On reserved networks the misclassification rate of a 20 °C exceedance is 6.86% when the statistic is computed on recorded days, 1.97% when every year is reconstructed and not screened, 1.53% under length screening and **0.36%** under the screen. Because most station-years sit nowhere near the threshold, we restrict the judgement to years where the call is close.

Table 10. Exceedance misclassification, reserved panel.

subset	observed days	reconstruct all	length-screened	screened
all retained	6.86%	1.97%	1.53%	**0.36%**
truth within 1.0 °C of the threshold	16.84%	9.50%	7.26%	**1.92%**
truth within 0.5 °C of the threshold	19.10%	13.12%	10.24%	**3.38%**

The advantage survives the restriction and is proportionally larger where the call is hard — a 67% reduction against length screening within half a degree of the threshold. The result is not an artefact of the threshold: swept from 14 °C to 24 °C, the screen's misclassification stays between 0.18% and 0.37% while the length rule stays between 0.84% and 1.53% (Figure 12c). The bias in the count of years above 20 °C falls from −0.76 years per station under recorded days and −0.32 under length screening to +0.06.

The 20 °C threshold is a modelling choice and is swept for that reason; the regulatory literature uses a seven-day average of daily maxima with criteria that differ by several degrees between jurisdictions and designated uses — 16 °C for core cold-water salmonid habitat, 13 °C for salmonid spawning (WAC 173-201A-020) — which is why we also report the seven-day endpoint throughout and sweep the threshold across that range.

Screening is not the only action the calibrated object supports, and for a threshold judgement it is not the best one. Discarding a station-year is a blunt response to uncertainty about a single call. The alternative is to keep every year and decline only the judgement the interval cannot support.

Table 11. Two action spaces on the reserved panel.

policy	years judged	misclassification	missed exceedances
screen at the 0.5 °C tolerance	67.5%	0.36%	0.39%
abstain where the interval straddles the threshold	**92.7%**	0.42%	0.64%
both	65.7%	**0.08%**	0.12%

Keeping 93% of station-years rather than 68% at essentially the same error rate matters to anyone assembling a regional archive. The two action spaces answer different questions — screening decides what enters an archive, abstention decides what one judgement may say — and the same calibrated object serves both.

Multi-year trends. Theil–Sen slopes of the annual peak have RMSE 0.143 °C yr^{-1} on recorded days, 0.044 with reconstruction and no screening, 0.048 under length screening and 0.041 under the screen. The panel averages eleven complete years per station, which is too short to resolve a decadal trend; we report the slope error as a diagnostic of the reconstruction rather than as a trend estimate, and the only claim we make from it is that reconstructing at all is the dominant step.

5.12 RQ8 — Whom does the screen keep?

A rule that lowers the average error by discarding hard station-years has bought accuracy with representativeness. §5.5 asked whether basin attributes predict how much the screen *gains* and found that none does. That is not the question an agency asks. The question is which stations the screen *keeps*, and in this application the hard sites are not a random subset: they are high-elevation, steep, donor-poor catchments, which are the cold-water headwaters the summer endpoint exists to describe.

At 70% retention the calibrated screen's per-station retention correlates with site elevation at Spearman $\rho =$ **-0.26 ($p = 0.011$)** on development and **-0.27 ($p = 0.077$)** on the reserved panel. The incumbent length rule is elevation-neutral: $\rho = -0.05$ ($p = 0.60$) and -0.10 ($p = 0.52$). On the reserved panel the screen retains **77.7%** of station-years in the lowest elevation tertile and **66.9%** in the highest, a spread of 10.8 points, against 1.7 points for the length rule. By donor regime the same pattern appears on development cases: the screen keeps 57.1% of station-years at donor-less stations and 72.5% at donor-rich ones, a 15.4-point spread, while the length rule tilts the other way. Dispersion of per-station retention roughly doubles, from 0.075 to 0.149.

Table 12. Error against representativeness, reserved panel at 70% retention (Figure 9).

rule	consequential-error rate	ρ(retention, elevation)	donor-regime retention spread	per-station retention SD
outage-length rule	8.81%	−0.10	4.5 pts	0.075
peak-overlap rule	3.75%	+0.16	2.4 pts	0.104
calibrated screen, global threshold	**2.53%**	**−0.27**	4.3 pts	0.149
calibrated screen, balanced within donor regime	2.52%	−0.27	**0.1 pts**	0.148
calibrated screen, balanced within elevation tertile	2.85%	**+0.08**	4.6 pts	0.148
calibrated screen, balanced within both	2.84%	+0.07	**0.1 pts**	0.144

The fix is cheap and it should be the default. Applying the screen's threshold *within* stratum rather than globally costs **0.31 points** of consequential-error rate on development cases and 0.30 on the reserved panel, and it removes both distortions: retention becomes flat across elevation tertiles and across donor regimes by construction. Stratifying on donor regime alone does not fix the elevation tilt, and stratifying on elevation alone does not fix the donor tilt, so an agency that cares about both must declare both. We therefore recommend that a screening rule for a public archive be specified with its strata, and that retention by catchment attribute be reported alongside the error rate, because a rule that silently removes the mountain stations from a national thermal archive has done damage that no error rate will show.

5.13 RQ9 — Does the result survive its assumptions?

Is missingness informative? Every case in this study transplants a mask onto a year whose values were observed, which is valid only if the mask is independent of the values it hides. Loggers are lost to floods, buried by ice and fouled in low flow, and all three coincide with anomalous conditions. Across the 6,472 real gaps in the archive, the mean anomaly of the seven days preceding a gap, relative to the station's own day-of-year climatology, is +0.018 °C for gaps of a week or less ($p = 0.44$), +0.038 °C for 8–29 days ($p = 0.56$), −0.002 °C for 30–89 days ($p = 0.98$) and −0.215 °C for 90 days and longer ($p = 0.081$). Donors recording through a gap run +0.063 °C warm during short gaps, which is detectable at this sample size ($p = 0.017$) and is 3% of the within-station anomaly scale and an eighth of a tenth of the tolerance. The design's exchangeability is defensible on magnitude rather than on a null result, and the one class that warrants a limitation is outages of three months

and longer.

Instrument error in the reference. Perturbing the reference series by 0.15 °C of normal noise — the upper end of logger accuracy — moves the development contrast from 6.32 to 6.21 points and the reserved contrast from 7.06 to 6.67 points, and the mean absolute error from 0.2416 to 0.2418 °C (Figure 12b). The conclusion is not inside the measurement error.

Estimators that may see only prior years. Under the stricter protocol in which only years before the outage are available, the panel falls to 33,633 development cases on 3,337 station-years, reconstruction error rises from 0.242 °C to 0.270 °C, and the contrast moves from −6.35 (−7.12, −5.52) to **−5.92 points (−6.72, −4.99)**, with coverage 0.906.

Transfer across agencies. Under leave-one-provider-out inside the development pool, the contrast is negative and its interval excludes zero for **all nine agencies**, ranging from −11.6 points (Finland) to −1.2 (Switzerland). Conformal coverage under provider transfer stays between 0.82 and 0.96.

Is the uncertainty honest? On the reserved networks the nominal 90% interval covers **90.9%** of cases marginally, within the pre-declared band of 0.87–0.93, and 86.3% to 94.9% within every pre-declared stratum (Figure 10a). Median widths adapt from 0.06 °C when the peak window is untouched to 1.87 °C when most of it is hidden. Those strata pool across networks, and conformal validity assumes an exchangeability that fails between river networks. Coverage inside individual reserved networks has mean 0.904 and median 0.913 but a minimum of **0.778**, with five of 36 networks below 0.85. The pooled interval is an honest statement about a population of networks and is **not** a guarantee at a single new site. Calibrating on the per-network conformal quantiles at the 0.9 level puts no network below 0.85 for a 1.5-fold widening, and **the screening result is identical under every calibration** to four decimal places, because the rank order the screen uses is a monotone transform of the bound and a change of scale cannot reorder it. A reader who distrusts the intervals need not distrust the frontier.

6. Discussion

What an agency should change, and in what order. Figure 14 is the procedure, end to end. The first recommendation is not to adopt a model and not even to change a screening rule. It is to look at the maintenance calendar. Holding the real distribution of outage lengths fixed and changing only when outages begin, a calendar concentrated in March to July would raise the consequential-error rate of the annual peak from 8.7% to **13.2%**, and one concentrated in September to January would lower it to **4.9%**. Those are swings of +4.4 and −3.9 percentage points, against a screening gain of 6.3 points at 70% retention — the same order of magnitude, and available before any data are lost rather than after. Our archives happen to sit near the good end of that range for outages of three months and more, and near the indifferent middle for everything shorter, so the headroom is real. Servicing loggers, replacing batteries and scheduling site visits in autumn converts recoverable outages into harmless ones; doing it in late spring does the reverse. Redundancy follows the same logic: our donor experiments show that a

second station is worth having in proportion to its synchronous anomaly correlation with the target rather than its mere existence, and our catchment analysis shows that synchrony is lowest at high-elevation, steep sites, which is where a redundant gauge buys the most.

The second recommendation is to change the axis, and the third is to stratify the change. An agency that screens station-years for warm-season extremes on maximum gap length should screen instead on how much of the station's climatological warm window the outage hides — widened, per §5.3, to 60 to 90 days rather than the 30 days of the metric itself, because one window of that width serves every warm-season endpoint and absorbs the year-to-year wander of the peak's position. On our reserved panel that substitution takes the consequential-error rate from 9.7% to 4.2% at equal retention with three variables and to 2.5% with the calibrated screen. But the threshold should be applied within strata, not globally, or the archive quietly loses its mountain stations; §5.12 prices that at a third of a percentage point. Equally, agencies should **keep** their length-based rules for annual means and cold extremes, and for the sine-fit parameters the standing guidance was written for, where our evidence says completeness is the better axis.

Why the standing guidance saw an inconsistency. Johnson et al. reported that the timing of missing data affected thermal-signal estimates inconsistently in space and time. Their three estimands have three different supports — a constant for the fitted mean, two lobes at the extremes for the amplitude, two lobes at the transitions for the phase — and only one of the three has a support narrow enough for a timing effect to appear. Observing all three at once, inconsistency is the only thing one could see. Our result does not contradict the observation; it decomposes it, and it predicts in advance which of their parameters should show a timing effect.

What generalises beyond rivers. The law is a statement about estimators and estimands, and its ingredients — an influence function and a mask — exist wherever a derived statistic is computed from an incomplete record. What does not generalise is the numerical crossover at 91 days: that depends on how concentrated a river's thermal year is and on how far the peak wanders between years, both of which are properties of this climate and this variable. What we would expect to transfer is the shape of the argument and the measurement protocol: locate the estimand's support, measure how far its location moves between periods, dilate by that amount, and condition on the result.

On uncertainty for derived statistics. The field reports daily prediction skill and almost never reports uncertainty on the annual statistic that is actually used. The two are not interchangeable for an extreme: a model with excellent daily RMSE can be badly wrong about a 30-day maximum if its errors are correlated within the peak window. Calibrated intervals on the derived statistic are cheap to produce once the problem is posed this way, and §5.13 shows they can be made honest either for a population of sites or for a single site, with the price stated.

7. Limitations

The operational benefit depends on the outage regime: under the geometry these networks actually experience the reserved-panel interval includes zero (§5.9), and

the headline contrast is computed on a mixture that includes a designed arm. The deployment analysis in §5.10 decides real incomplete records but cannot score them; its risk numbers are a labelled extrapolation from scored cases. The archive-level consequence of changing rules is small — the choice of rule moves a regional mean annual peak by less than 0.1 °C — and we do not claim otherwise. The pooled conformal interval is not valid at a single new site unless the cluster calibration is used, and that costs width. The screen uses a station's own history, so nothing here applies to ungauged sites. Outages that straddle a calendar year are outside the design, and the law implies they should be nearly harmless for both peaks and costly for both means — a prediction we have not tested. Fully observed calendar years are, by selection, the better-instrumented part of an archive. Catchment attribution and the representativeness analysis are restricted to the United States subset where GAGES-II applies, and 140 of 204 stations match. Missingness is weakly informative and the transplant design is defended on the magnitude of that effect rather than on its absence (§5.13); the defence is weakest for outages of 90 days and longer. The primary tolerance is a modelling choice at 0.47 interannual standard deviations rather than an ecological threshold, which is why the frontier is reported at 0.25, 0.5 and 1.0 °C and the exceedance threshold is swept from 14 °C to 24 °C. Eleven complete years per station is too short for a decadal trend claim, and none is made.

8. Conclusion

Whether a data gap matters depends on how much the statistic being computed depends on the days it hides. Measuring that share — the hidden mass on the estimand's support — orders the regimes that a completeness threshold cannot see, and it does so continuously: sweeping the support of a warm-season mean from one day to a full year, the advantage of a peak-window axis over a completeness axis decays from +0.074 AUC and crosses zero at 91 days. Because a support has to be located before it can be conditioned on, and a river's hottest window wanders by one to two weeks between years, the conditioning set that works is the support widened by that wander; one climatological window of 60 to 90 days is near-optimal for every warm-season endpoint we examined. On the estimands the standing guidance was written for, that same law says the guidance is right, which is why the correction is to the transfer of a threshold rather than to the threshold.

Acting on it is cheap. Screening on the governing axis, pre-registered and scored once on 36 networks reserved before any analysis, reduces the consequential-error rate by 7.1 percentage points relative to published guidance at equal retention and reduces severe underestimation of the summer peak sevenfold; most of that is obtained by three variables in a logistic regression. Taken to the 2,422 real incomplete station-years the study could not score, the rule reverses the admission decision for one in six of them and removes the 17 station-years that published guidance admits with the annual peak never observed.

Two things should be said as loudly. The gain is regime-dependent, and under the outage calendar these agencies currently keep it is not resolvable — because long outages already begin in autumn, which is itself the most actionable finding here: moving maintenance into spring would cost more than the screening rule saves. And a screen that selects on difficulty selects against high, steep, donor-poor catchments

unless its threshold is applied within strata, which costs a third of a percentage point and should be standard. Keep length-based completeness rules for averages and for sine-fit regime parameters; replace them with support-based rules for warm-season extremes; and schedule the loggers.

Figures

Figures 1, 2, 3 and 14 are schematics, drawn in draw.io; their editable sources are in `figures/v17/drawio/` and the PDFs are exported from those sources by `make figures-drawio`. Figures 4, 6, 10, 11, 12 and 13 are produced from the analysis artifacts by `scripts/188_v17_figures.py` and `scripts/204_v17_figures_q1.py`; Figures 5, 7, 8 and 9 by `scripts/220_v18_figures.py`. Figures marked SI are supporting information.

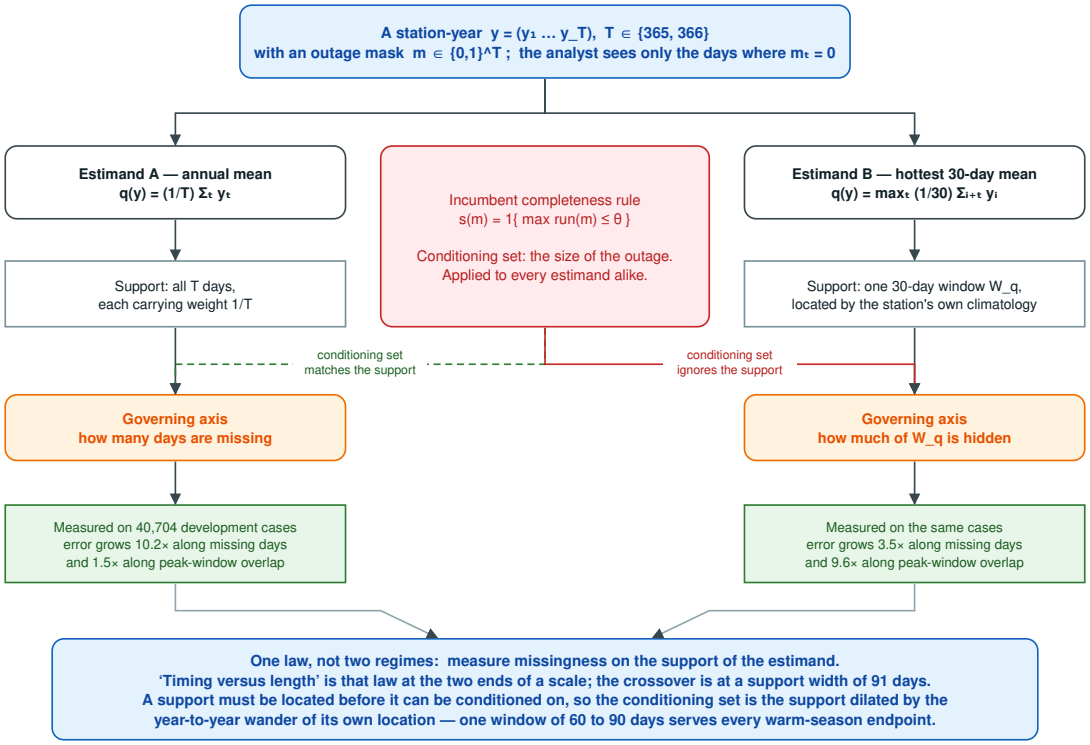


Figure 1. Why one completeness threshold cannot serve two estimands: the same station-year and outage mask feed an annual mean and a hottest 30-day mean, the two have different supports and therefore different governing axes, and the measured error gradients along each are quoted.

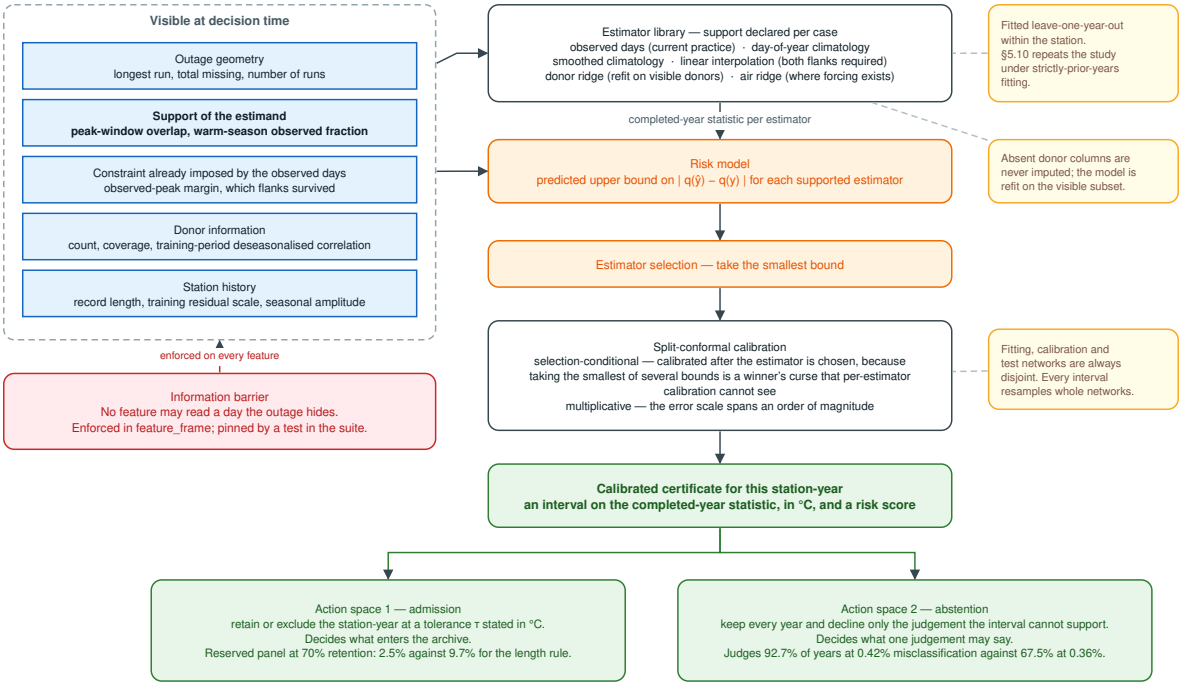


Figure 2. The screening system: visible features under an information barrier, the estimator library with per-case support, the risk model, selection-conditional multiplicative conformal calibration, and the two action spaces.

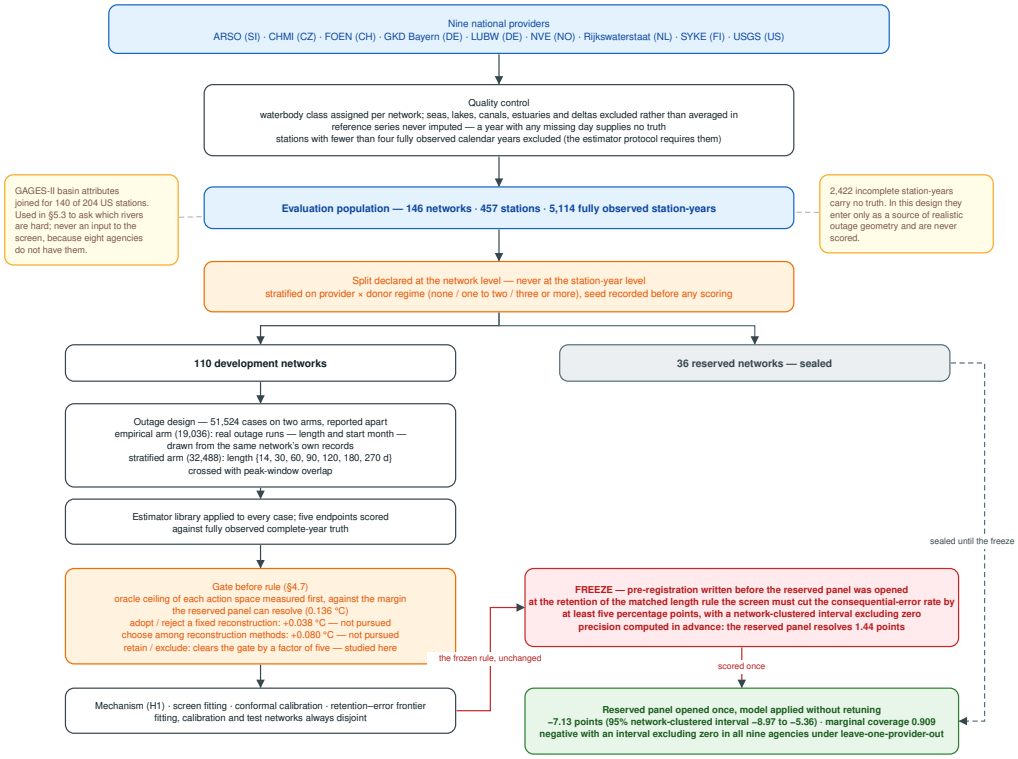


Figure 3. Study design and data flow, from nine providers to the single opening of the reserved panel, with the quality-control exclusions, the network-level stratified split, the §4.7 gate and the pre-registered freeze in execution order.

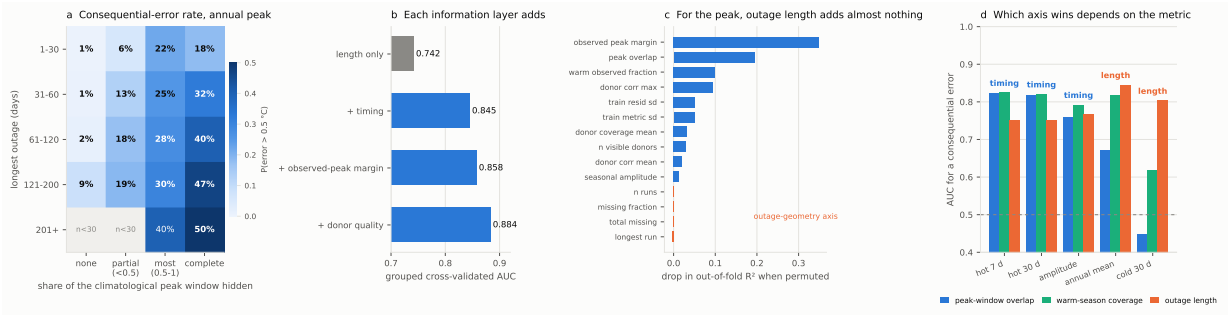


Figure 4. Mechanism: the length-by-overlap reversal (a), nested feature sets (b), permutation importance (c), endpoint dependence (d).

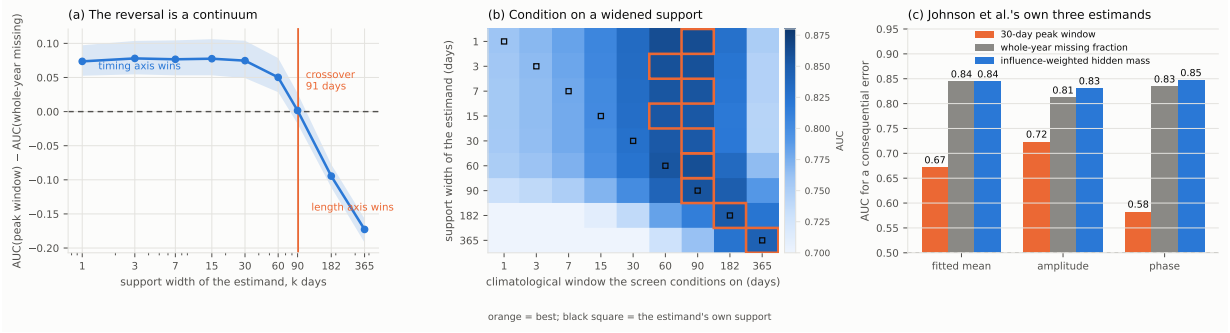


Figure 5. The law. (a) The advantage of a fixed 30-day peak window over whole-year completeness against the support width of the estimand, with network-clustered intervals and the crossover at 91 days. (b) AUC for every estimand against every conditioning window; the best window is 60–90 days for every estimand narrower than that, and the black square marks the estimand's own support. (c) The three sine-fit parameters Johnson et al. use, by axis.

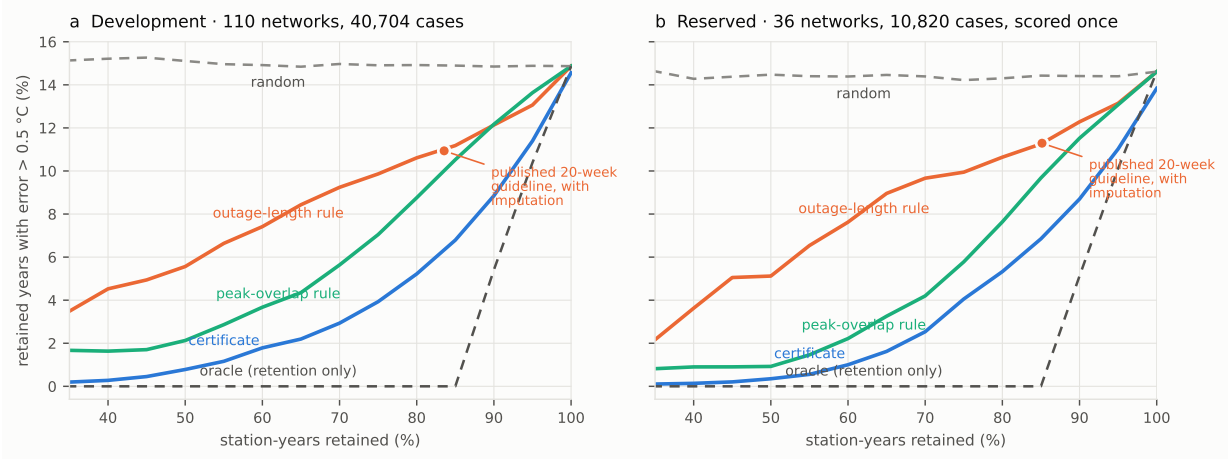


Figure 6. Retention–error frontier on development cases (a) and on the reserved panel (b).

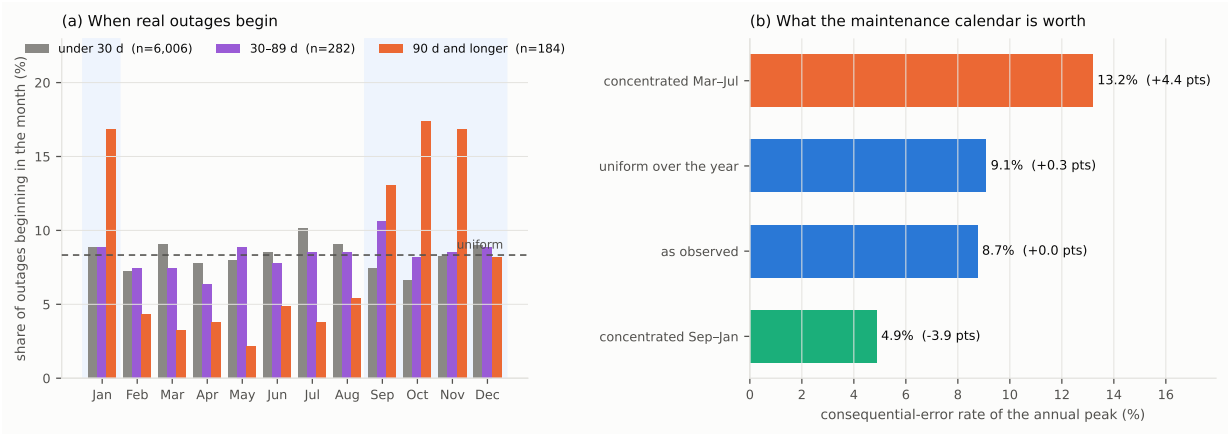


Figure 7. (a) Start month of 6,472 real interior gaps by length class; outages of 90 days and longer begin in autumn and winter. (b) What the maintenance calendar is worth: the consequential-error rate under four placement calendars, holding the real outage-length distribution fixed.

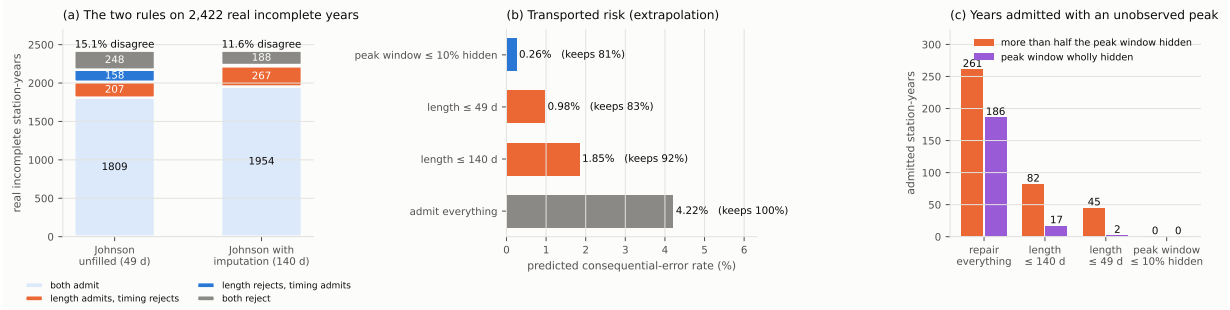


Figure 8. The rules on 2,422 real incomplete station-years. (a) Agreement and disagreement at each published limit. (b) Transported risk of each admitted set. (c) Station-years admitted with the peak window more than half, or wholly, unobserved.

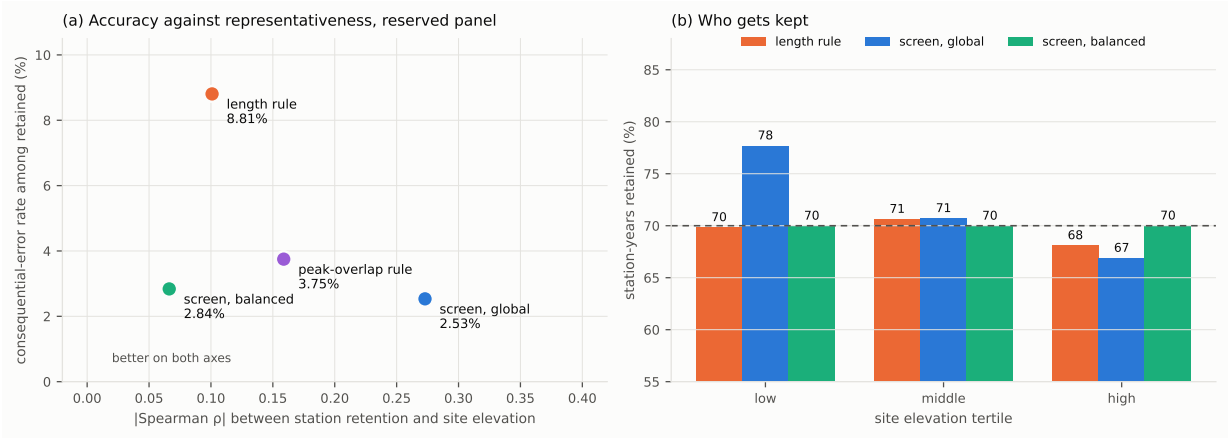


Figure 9. (a) Consequential-error rate against the correlation between station retention and site elevation, reserved panel. (b) Retention by elevation tertile under three rules.

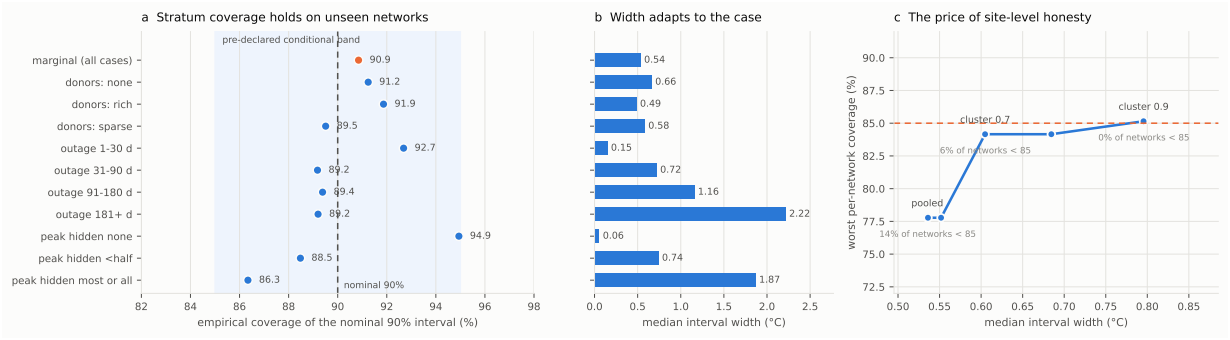


Figure 10. Conformal coverage by stratum (a), interval width against peak-window overlap (b), and the price of site-level validity (c).

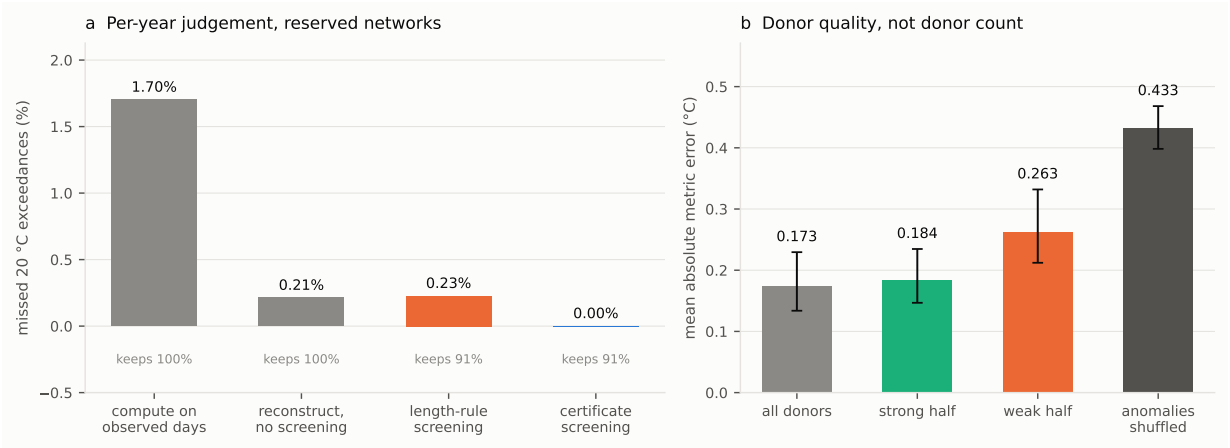


Figure 11. Multi-year trend error (a) and the donor controls (b).

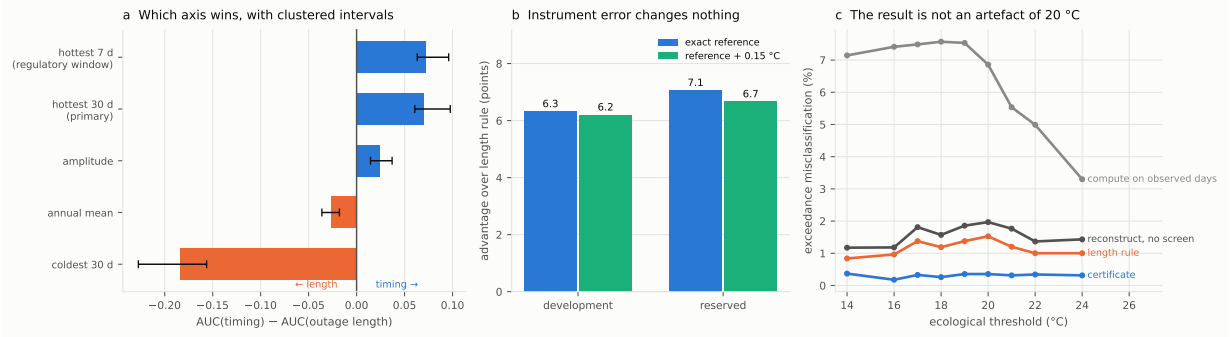


Figure 12. Clustered intervals on the axis contrasts (a), reference-noise perturbation (b), exceedance-threshold sweep (c).

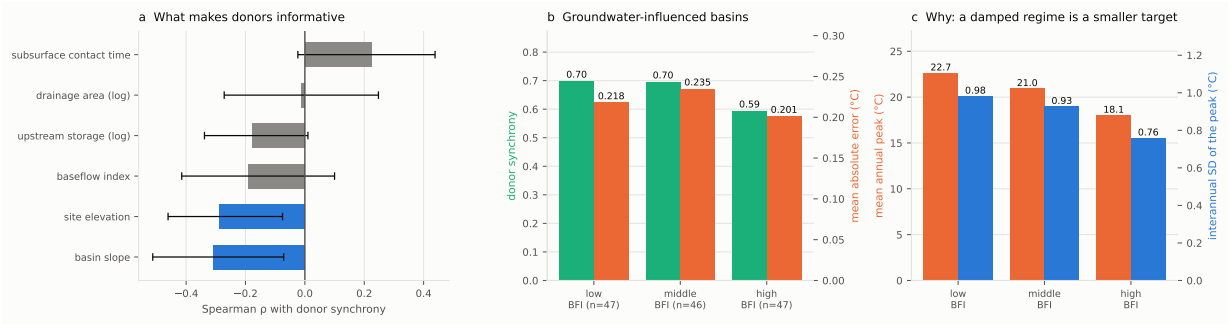


Figure 13. Catchment attribution on the United States subset, and the decomposition of the baseflow effect (c).

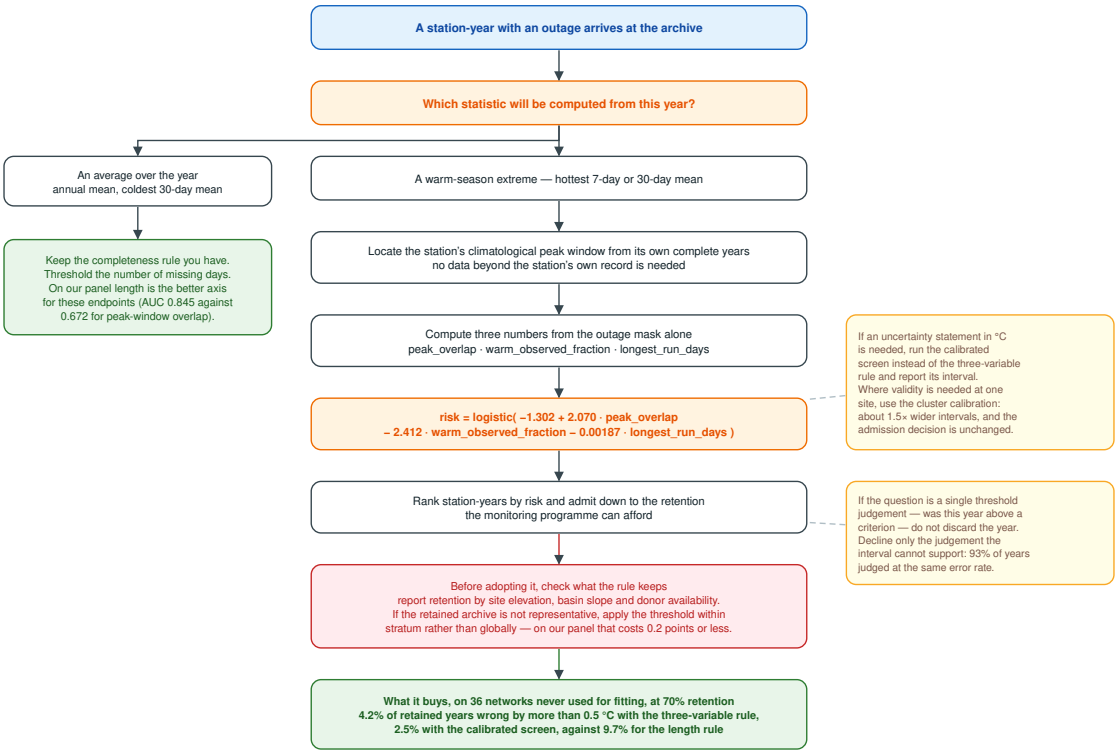


Figure 14. What an agency does with the result, including the representativeness guardrail.

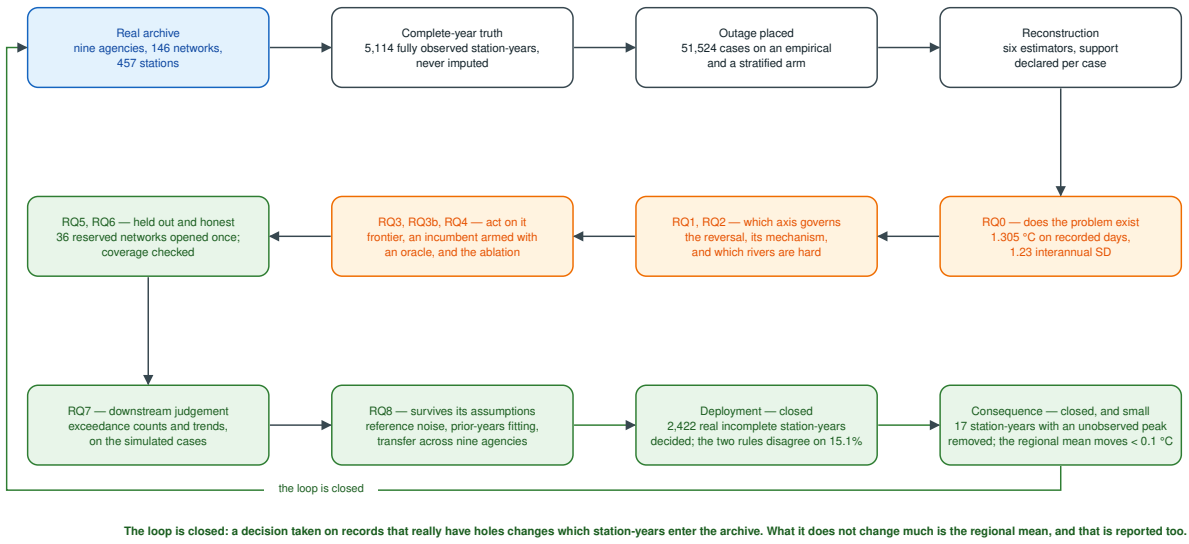


Figure SI-1. The evidence chain by research question, and which links this version closes.

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Data and code availability

Daily temperature records are from the nine public providers listed in the supporting information. Station coordinates are from those providers' own catalogues, the USGS NWIS site service, and ARSO's public station feed; daily air temperature is from NASA POWER. Basin characteristics are from GAGES-II (Falcone, 2011). All analysis code, artifacts, figures, draw.io figure sources and the frozen pre-registration are in the accompanying repository; `results/v17_fitness/evidence_ledger.csv` and `paper/v18/claims_v18.md` map each claim to the script, artifact and digest behind it, and every table in this article is re-derived from those artifacts by an automated audit that runs in the test suite.